

Non-Linear Finite Element Methods

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Technical Documentation
Summer Semester 2023

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1 Implemented Theory:

We are considering a 2-dimensional plate with a hole subjected to uniform farfield tractions. The goal is to investigate the behavior of the plate using finite element analysis (FEM). Due to the symmetries of the problem, only a quarter of the plate needs to be considered.

1.1 Weak Formulation

The weak form is obtained by multiplying the governing equations by virtual displacements and integrating over the domain. The Galerkin's weak form, also known as the generalized weak form is expressed as:

$$\delta W^e = \delta \underline{\hat{u}}^{eT} \left[\left(\int_{\Omega_{\square}} \rho \underline{N}^T \cdot \underline{N} \det(\underline{J}) dV_{\xi} \right) \cdot \ddot{\underline{u}}_e + \int_{\Omega_{\square}} \underline{B}^T \cdot \underline{\sigma} \det(\underline{J}) dV_{\xi} - \left(\int_{\partial\Omega_e} \underline{N}^T \cdot \underline{\hat{t}} dS + \int_{\Omega_{\square}} \underline{N}^T \cdot \underline{b} \det(\underline{J}) dV_{\xi} \right) \right]$$

For the current problem with quasi-static conditions and no body forces, the reduced weak form is:

$$\delta W^e = \delta \underline{\hat{u}}^{eT} \left[\int_{\Omega_{\square}} \underline{B}^T \cdot \underline{\sigma} \det(\underline{J}) dV_{\xi} - \left(\int_{\partial\Omega_{\square}} \underline{N}^T \cdot \underline{\hat{t}} \det(\underline{J}) dS_{\xi} \right) \right]$$

Here, δW^e represents the virtual work, $\delta \underline{\hat{u}}^e$ is the virtual displacement vector, $\underline{B}$ is the strain-displacement matrix, $\underline{\sigma}$ is the stress vector, $\underline{J}$ is the Jacobian matrix, $\underline{N}$ is the shape function matrix, $\underline{\hat{t}}$ is the traction vector, and $\Omega_{\square}$ and $\partial\Omega_{\square}$ represent the problem domain and its boundary, respectively.

To discretize the problem, Galerkin's approach is employed, which introduces test functions to satisfy the system conditions. The Jacobian is calculated to interpolate the equations over the entire element by taking the derivative of the test function with respect to the unit coordinates, which range from -1 to 1. The Jacobian is also used for the conversion of the Bridge "B" matrix. For numerical integration, Gauss quadrature points and weights are utilized.

1.2 Tangential Stiffness Matrix

The tangent stiffness matrix, denoted as K_t^e , is an important component in finite element analysis. It relates the change in internal forces to the change in displacements and is used to solve the system of equations. In this case, the tangent stiffness matrix is given by:

$$K_t^e = \underline{B}^T \cdot \underline{C}_t \cdot \underline{B} \cdot \det(\underline{J}) \cdot \text{weights}$$

Here, weights represents the Gauss quadrature weights, $\underline{B}$ is the strain-displacement matrix, $\underline{C}_t$ is the material's tangent stiffness matrix, and $\det(\underline{J})$ is the determinant of the Jacobian matrix.

1.3 Internal Force Vector

The internal force vector, $\hat{F}_{\text{int}}$, represents the contribution of internal stresses to the equilibrium of the system. It is computed by integrating the product of the strain-displacement matrix, stress vector, Gauss quadrature weights, and the determinant of the Jacobian matrix over the domain:

$$\hat{F}_{\text{int}} = \int_{\Omega_{\square}} \underline{B}^T \cdot \underline{\sigma} \det(\underline{J}) dV_{\xi}$$

1.4 External Force Vector

The external force vector, $\hat{F}_{\text{ext}}$, accounts for the effects of tractions applied to the boundary of the domain. For this specific problem statement, it is computed by integrating the product of the shape function matrix, traction vector, Gauss quadrature weights, and the determinant of the Jacobian matrix over the boundary:

$$\hat{F}_{\text{ext}} = \int_{\partial\Omega_{\square}} \underline{\underline{N}}^T \cdot \hat{\underline{t}} \det(\underline{J}) dS_{\xi}$$

1.5 Residual

The residual vector, denoted as $\mathbf{R}$, represents the difference between the internal force vector ($\hat{F}_{\text{int}}$) and the external force vector ($\hat{F}_{\text{ext}}$). It is used to solve the system of equations and achieve equilibrium. The residual vector is given by:

$$\mathbf{R} = \hat{F}_{\text{int}} - \hat{F}_{\text{ext}}$$

1.6 Newton-Raphson Method

In the Newton-Raphson iteration scheme, the displacements ($\hat{u}^g$) are iteratively updated until a convergence criterion is met using the equation:

$$u^{k+1} = u^k + \Delta u$$

At each iteration, the tangent stiffness matrix (K_t^e) and the residual vector ($\mathbf{R}$) are computed and accumulated in K_{global} and R_{global} , and the displacement increment is obtained. Here, u^{k+1} is the updated displacements at the $k + 1$ iteration, u^k is the displacements at the current k iteration, and Δu is the displacement increment. The equation for the displacement increment:

$$K_{\text{global}} \cdot \Delta u = -R_{\text{global}}$$

where $\mathbf{K}_{\text{global}}$ is the global stiffness matrix and $\mathbf{R}_{\text{global}}$ is the global residual vector.

The iteration continues until a convergence criterion based on the norm of the residual vector is met. Once the change in displacements becomes sufficiently small, indicating convergence, the iteration is stopped, and the final solution is obtained.

1.7 Variant Information

For the specific problem variant mentioned, which involves a 'Tri' variant with a quad mesh divided into 2 triangular elements. We use 6 Gauss quadrature points for each element along with their respective weights. These quadrature points are employed for numerical integration in the calculations of the weak form, tangent stiffness matrix, internal force vector, and external force vector.

2 User Manual

This user manual provides instructions on how to use the code for analyzing the behavior of a 2D plate with a hole subjected to uniform farfield tractions using finite element analysis (FEM). The code is written in MATLAB.

2.1 Installation

Before using the code, ensure that MATLAB is installed on your computer. If MATLAB is not installed, download it from the official MathWorks website (www.mathworks.com) and follow the installation instructions.

Once MATLAB is installed, download all the required files, including the built-in files, `element_routine.m`, `main_routine.m`, `Elastic_plastic_material_model`, mesh information, and the `Assembly.m` function file. Save all the files to a directory of your choice. This directory will be called the "working directory" throughout this user manual.

2.2 Running the Code

To run the code and analyze the behavior of the 2D plate, follow these steps:

1. Download and save all the built-in files along with the `element_routine.m` and `main_routine.m` to the current working directory.
2. Open the files using MATLAB software.
3. Set the working directory to the directory where you saved the code files. You can set the working directory in MATLAB by using the "Current Folder" panel on the left side of the MATLAB interface.
4. Modify the assignment of parameters in the main routine of the program according to your specific needs:
 - Young's modulus (E): 210 [GPa]
 - Poisson's ratio (ν): 0.3
 - Initial yield stress (σ_Y): 0.2 [GPa] (For linear deformation, put $\sigma_Y = E$)
 - Linear, isotropic hardening modulus (`hard_iso`): 1.0 [GPa]
 - Increase in yield stress stemming from exponential hardening (`deltaY`): 0.0 [GPa]
 - Shape parameter for exponential hardening (`h_sh`): 15
 - Linear kinematic hardening modulus (`hard_kin`): 0.0 [GPa]
5. The above parameters are preset in the `main_routine.m` file. Changes can be made for user-specific requirements.
6. The code allows users to modify and choose the desired meshing profile by uncommenting the corresponding section. This provides flexibility in customizing the mesh information to meet specific requirements.
7. After assigning the inputs, the user can run the program in the `main_routine.m` file. You can do this by typing `main_routine` in the MATLAB command window and pressing Enter.
8. Once the analysis is complete, the program will generate various output files containing the stress and displacement distributions for different mesh types and orientations.

2.3 Interpreting the Results

The program generates output files that contain the stress graphs with different directions (0° , 90° , and 45°) for each mesh type (uniform, Coarse, Fine, and vFine). Additionally, displacement distributions are also generated for visualization. The output files are named according to the mesh type and orientation.

To interpret the results, you can open the output files using MATLAB or any other software that can read and visualize data files. The results are typically visualized as color maps, where different colors represent different stress or displacement values.

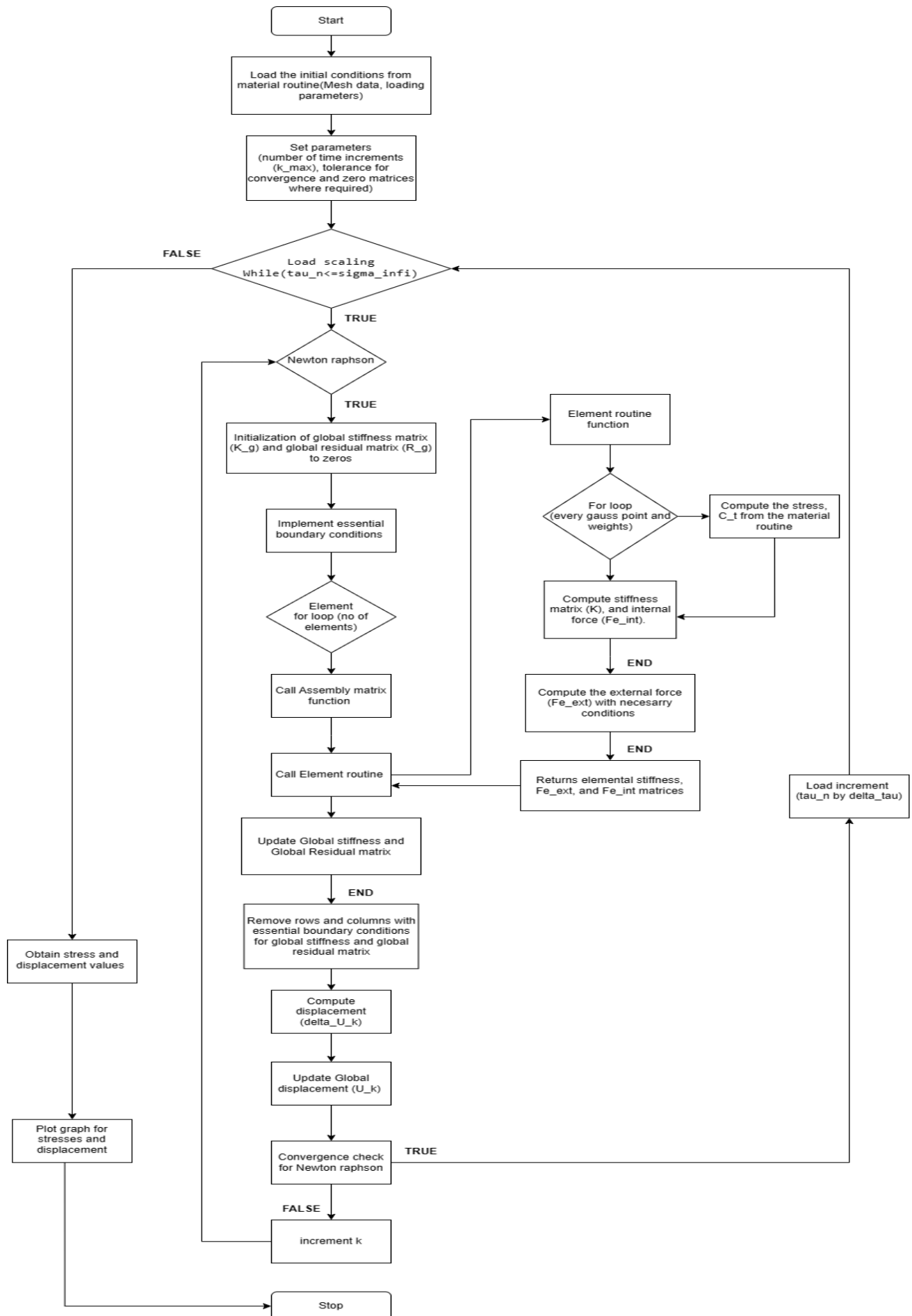
By analyzing the stress and displacement distributions, you can gain insights into the behavior of the 2D plate under uniform farfield tractions. You can observe how different mesh types and orientations affect the stress and displacement patterns.

2.4 Troubleshooting

If you encounter any issues while running the code or interpreting the results, consider the following troubleshooting steps:

- Double-check that you have set the working directory correctly in MATLAB. Make sure all the code files and input files are saved in the working directory.
- Ensure that you have modified the parameters in the Material Routine correctly according to your requirements.

3 Program structure



4 Convergence Study and Results

4.1 Linear Analysis

4.1.1 Uniform Mesh

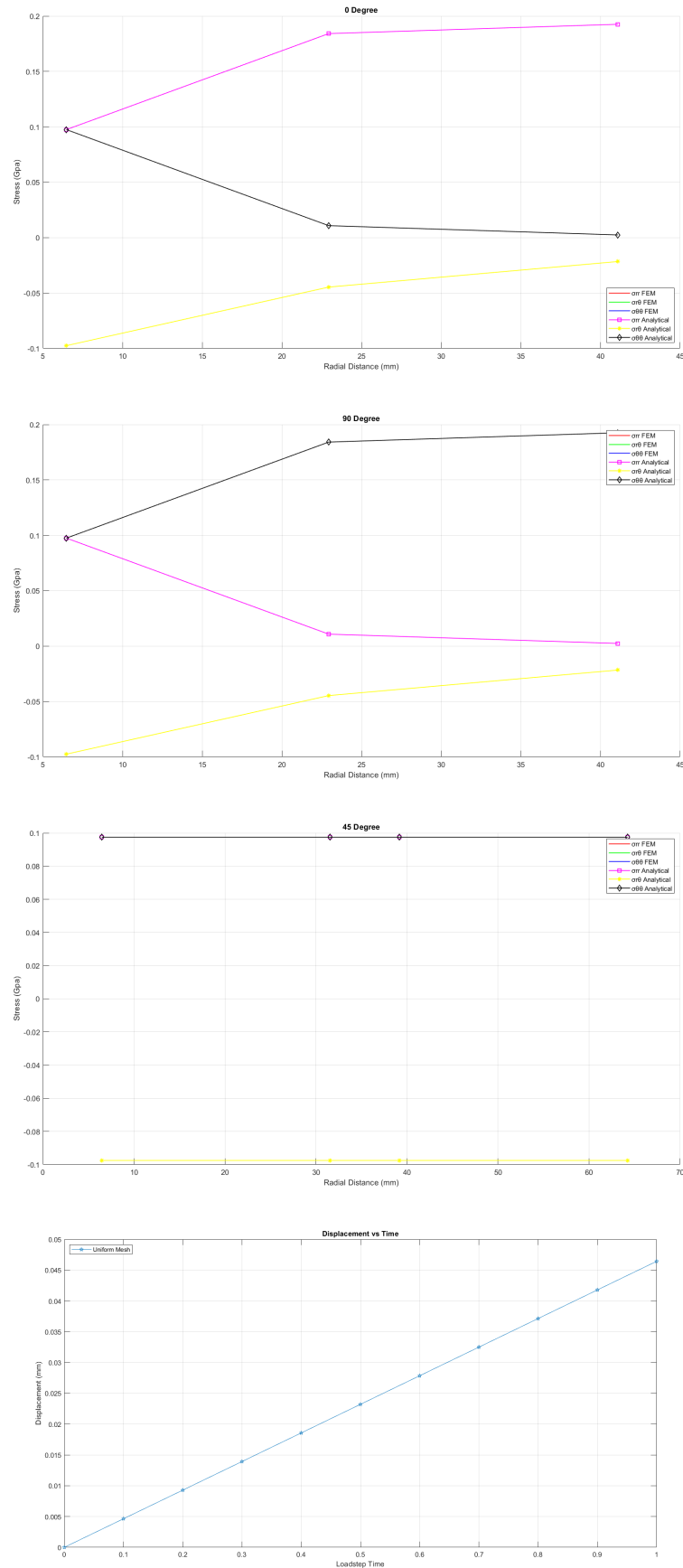


Figure 1: Stress distribution and Displacement in the 2D plate with a uniform mesh.

4.1.2 Coarse Mesh

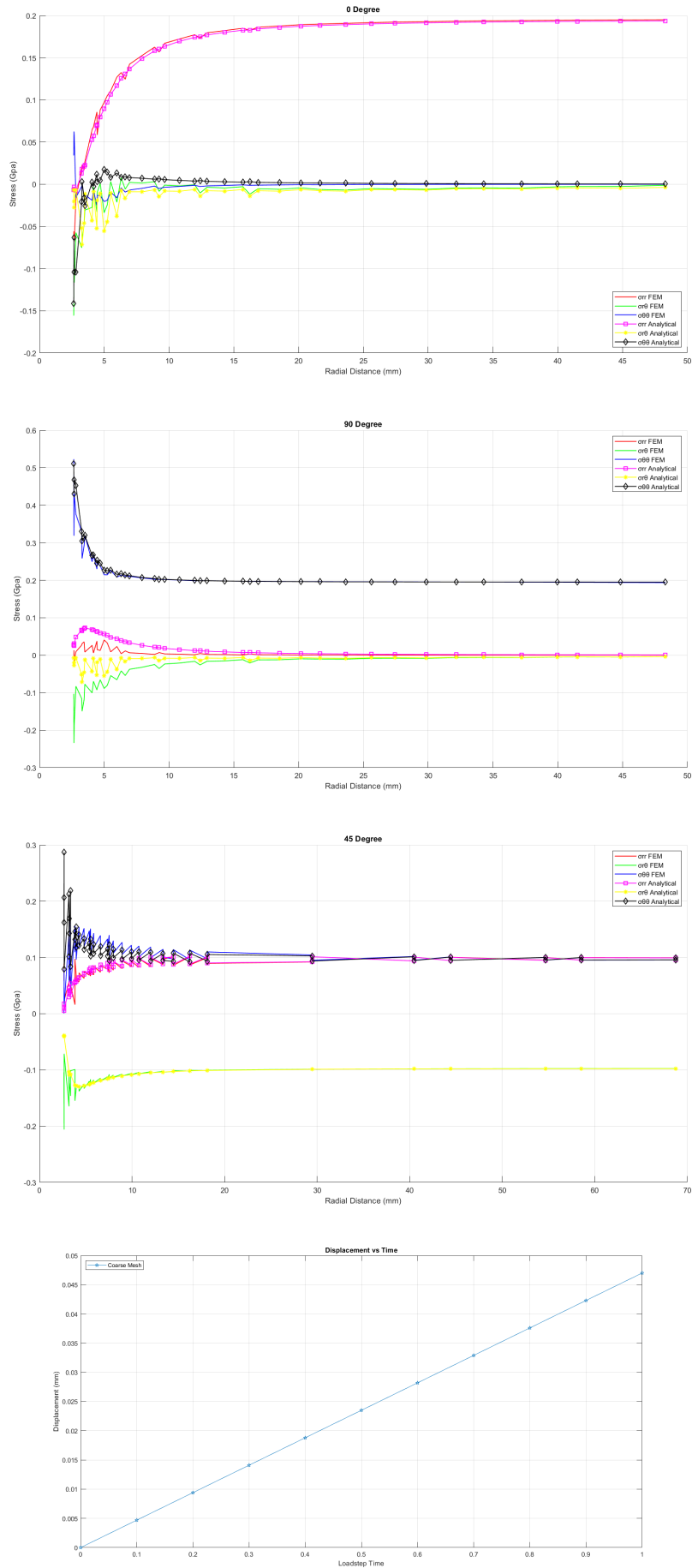


Figure 2: Stress distribution and Displacement in the 2D plate with a Coarse mesh.

4.1.3 Fine Mesh

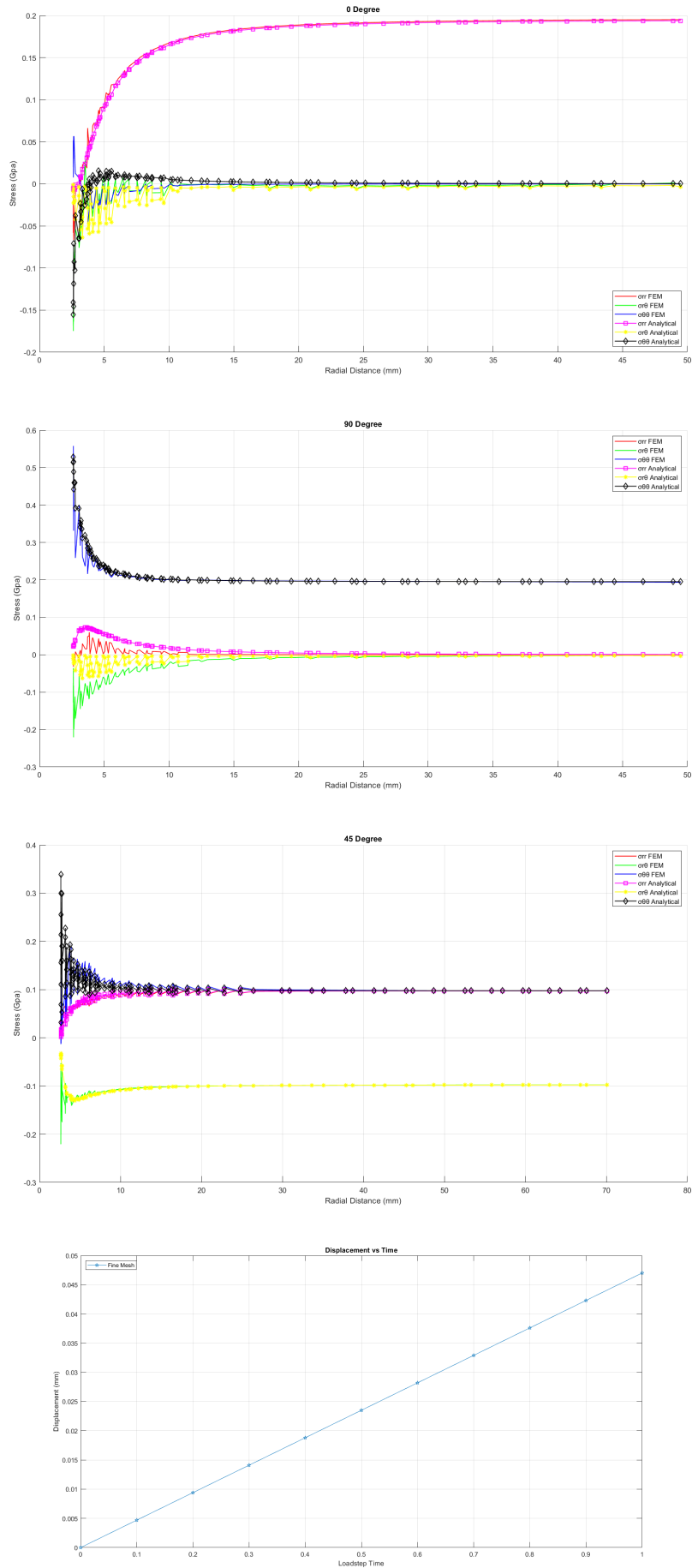


Figure 3: Stress distribution and Displacement in the 2D plate with a Fine mesh.

4.1.4 vFine Mesh

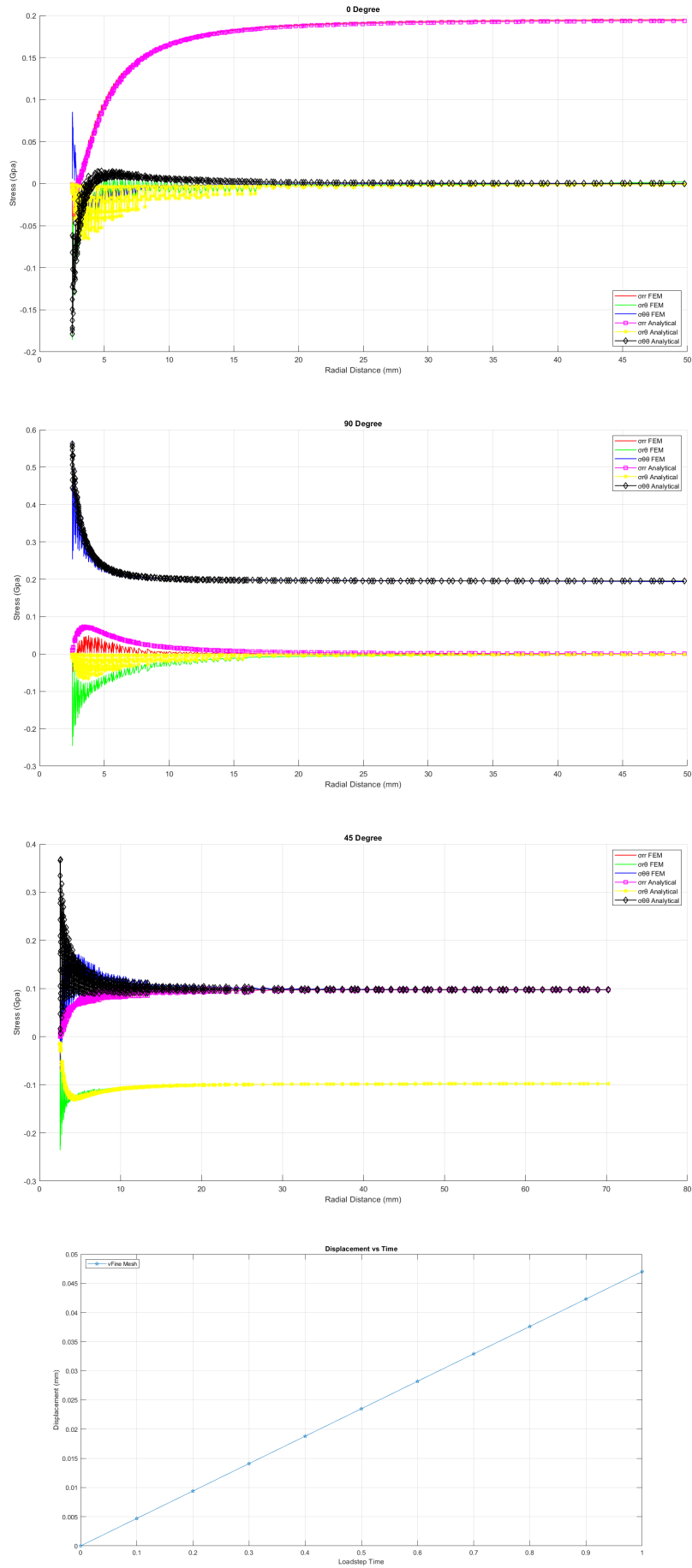


Figure 4: Stress distribution and Displacement in the 2D plate with a vFine mesh.

4.2 Nonlinear Analysis

4.2.1 Uniform Mesh

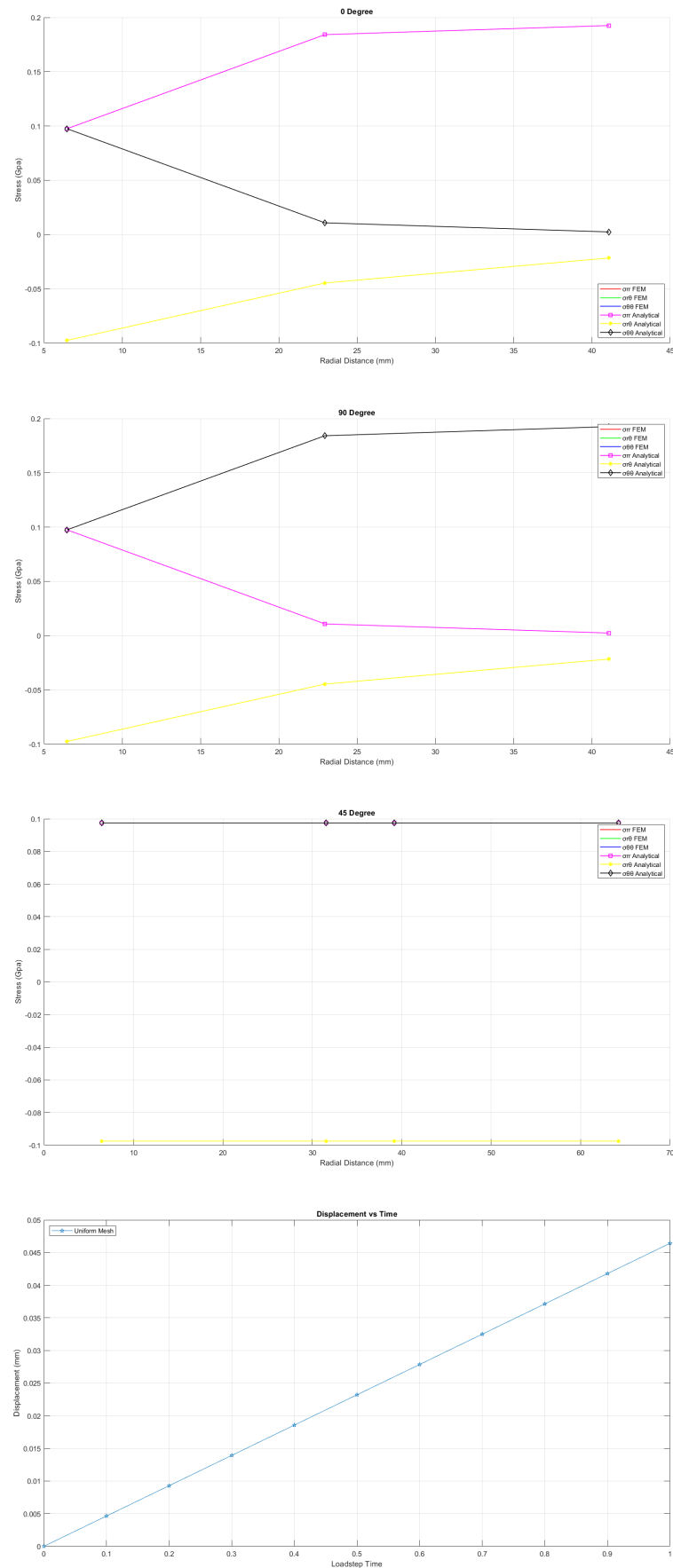


Figure 5: Stress distribution and Displacement in the 2D plate with a uniform mesh.

4.2.2 Coarse Mesh

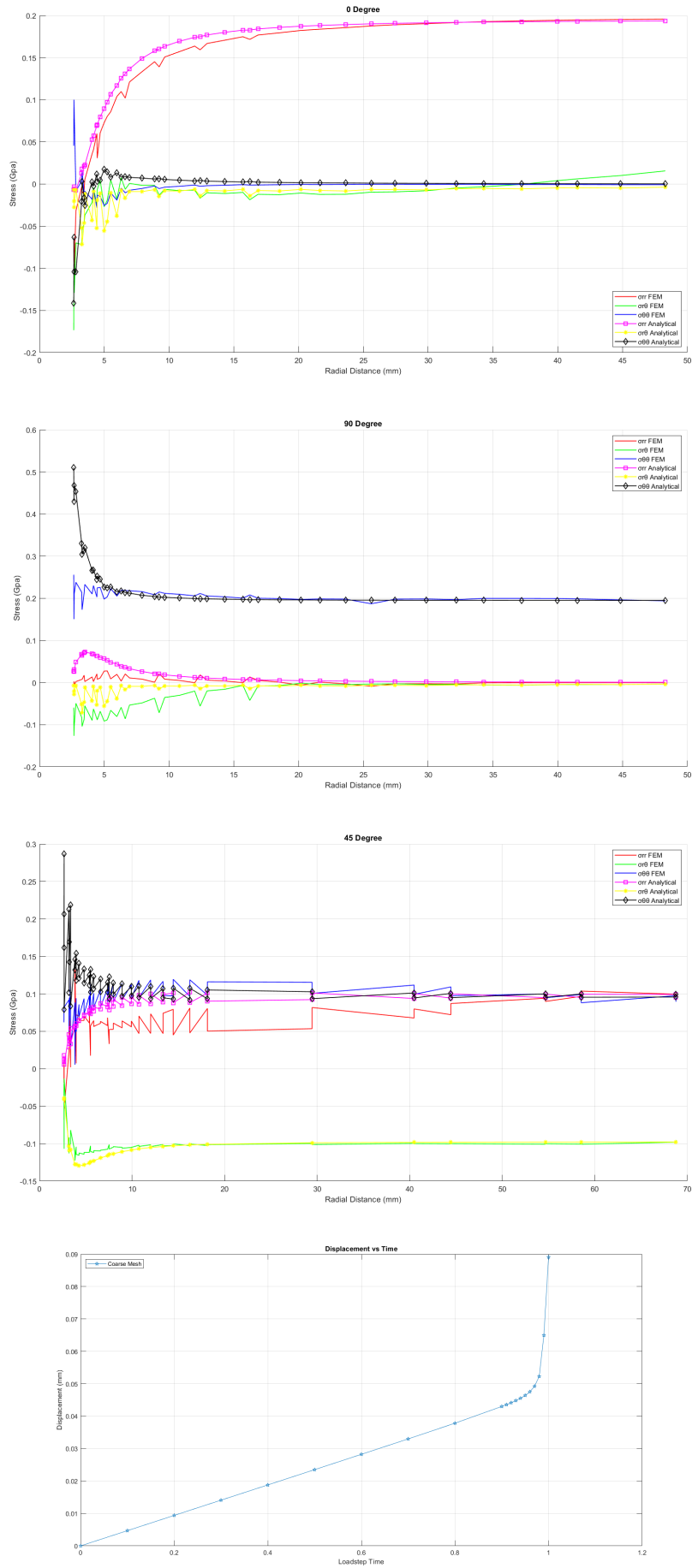


Figure 6: Stress distribution and Displacement in the 2D plate with a Coarse mesh.

4.2.3 Fine Mesh

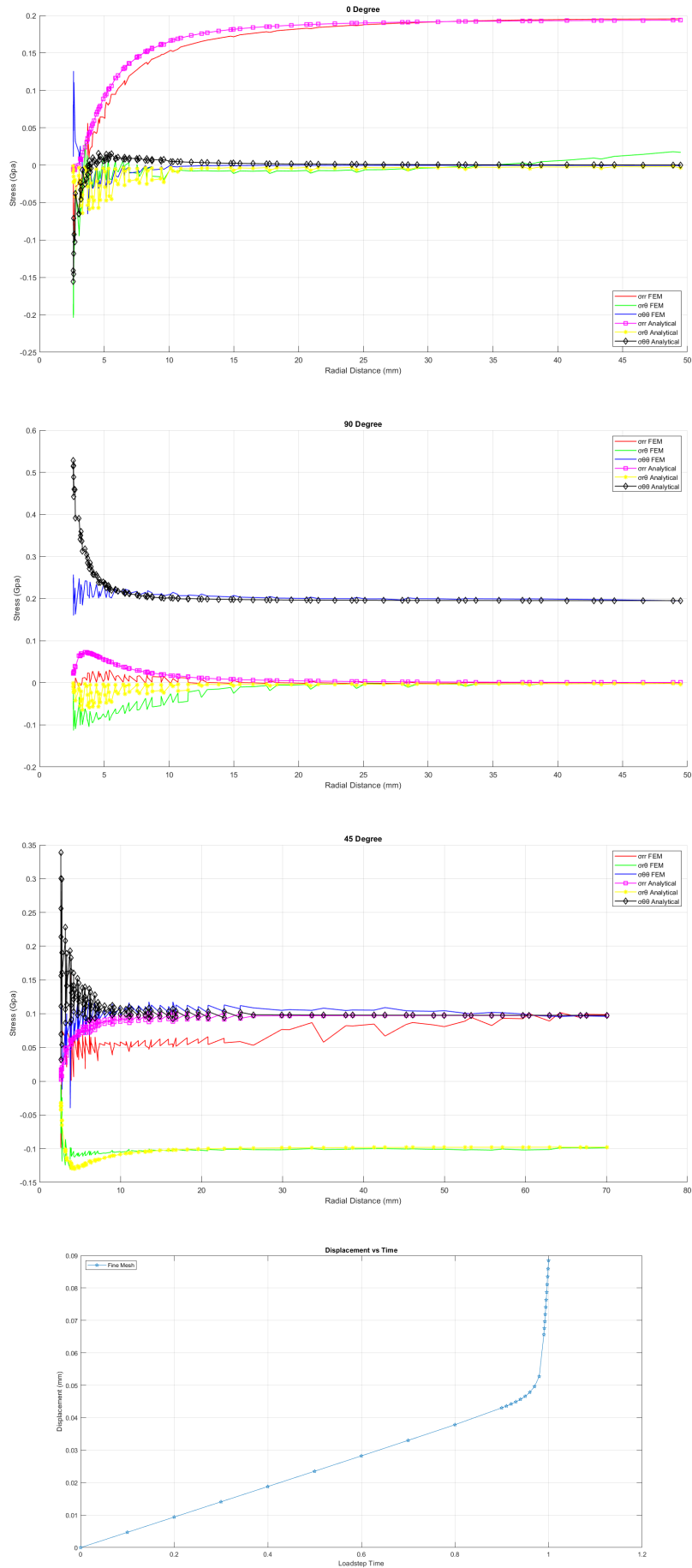


Figure 7: Stress distribution and Displacement in the 2D plate with a Fine mesh.

4.2.4 vFine Mesh

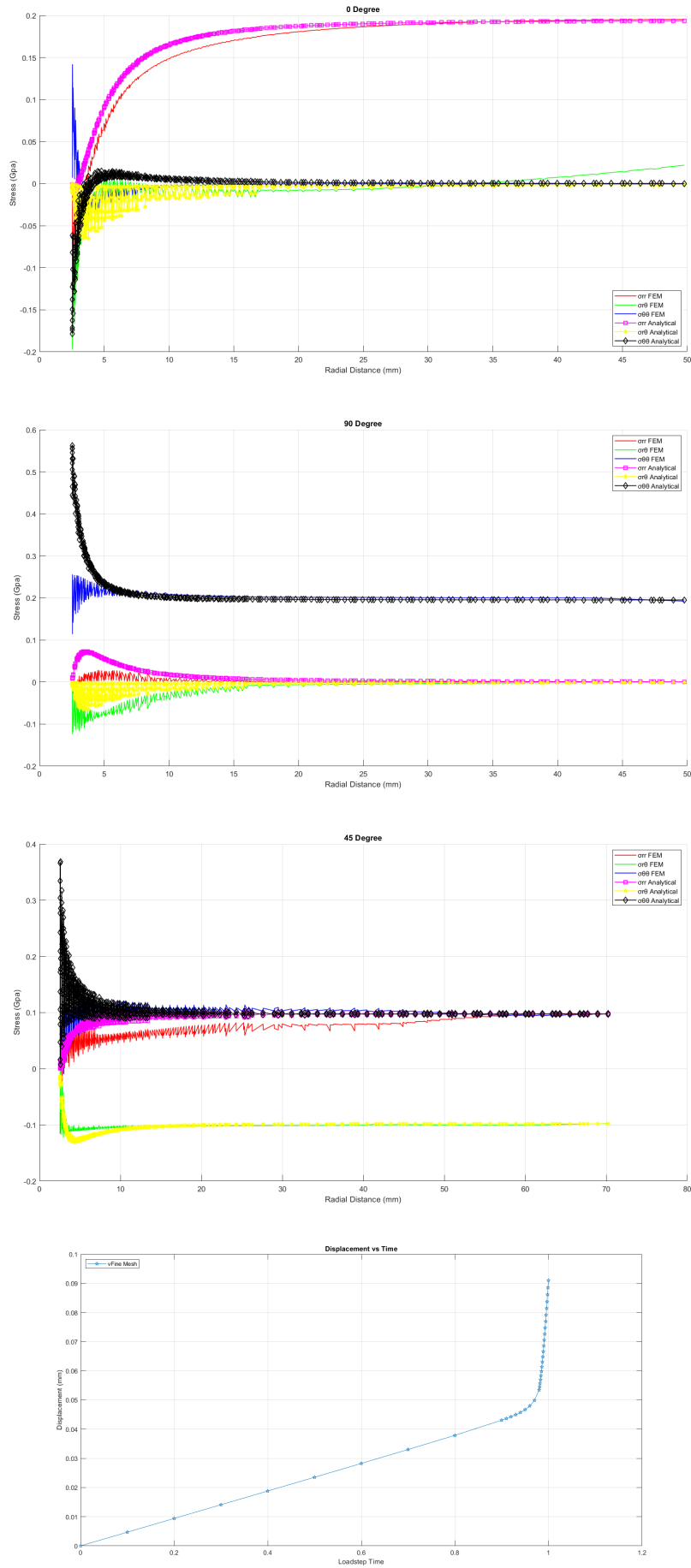


Figure 8: Stress distribution and Displacement in the 2D plate with a vFine mesh.